



12.4.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Unit 12, Lesson 5: Pythagorean Theorem and Tangent Lines



Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.



Learning Target

- ☐ I can use the Pythagorean Theorem to solve mathematical and real-world problems involving the length of a tangent segment to a circle.



12.5.1 Warm-Up: Ice Problem

1. An igloo, also known as a snow house or snow hut, is a type of shelter built out of snow.



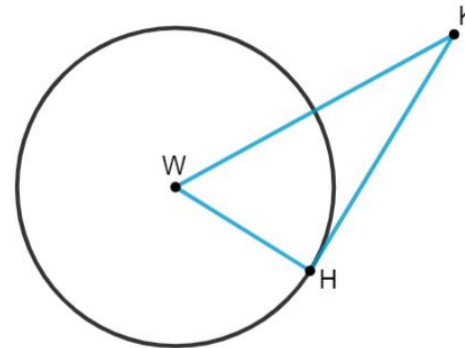
- a. What is the first question that comes to your mind?
- b. Write a question that would require someone to use geometry to solve.



12.5.2 Exploration: Pythagorean Theorem and Tangent Lines

Work with your partner to complete the following.

1. Circle W has point H on the circle. $\overline{WK}$ and tangent $\overline{KH}$ intersect at point K outside the circle.



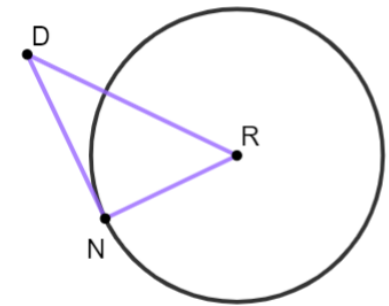
- a. Discuss with your partner what can be concluded from the given information. Justify your conclusions. Give at least two conclusions.

- b. Ask a classmate for two conclusions they drew. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials

- c. Discuss with your partner what information is needed to determine the length of $\overline{WK}$.
- d. Explain why any triangle drawn, where the radius and a tangent are the legs, will result in a right triangle.

2. Circle R is shown with point D outside the circle, where point N is on the circle. $DN = 10$, $NR = 8$, and $DR = 4\sqrt{10}$.



- a. Use the Pythagorean Theorem to determine if $\overline{DN}$ is a tangent to circle R .

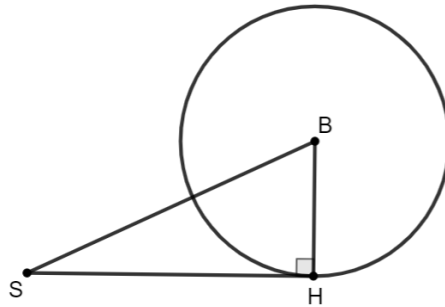
- b. Complete the statements.

$\triangle DRN$ ☐ is ☐ is not a right triangle. Therefore $\overline{DN}$ ☐ is ☐ is not a tangent line to circle R .



12.5.3 Guided Practice: Pythagorean Theorem and Tangent Lines

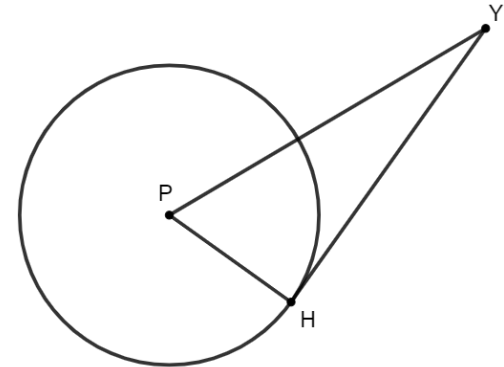
1. Circle B has point H on the circle and point S outside the circle. $m\angle SHB = 90^\circ$, $SB = 13$, and $BH = 5$.



- a. What is the length of $\overline{SH}$?
- b. What is the area of $\triangle SBH$?

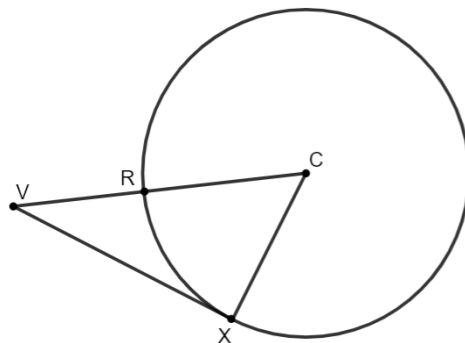
2. Circle P has point H on the circle and point Y outside the circle, where $\overline{PY}$ and tangent $\overline{HY}$ intersect. $PH = 10$ and $HY = 17$.

What is the length of $\overline{PY}$?



3. Circle C has points R and X on the circle and point V outside the circle, where $\overline{VC}$ and tangent $\overline{VX}$ intersect at point V . $VX = 24$ and $VR = 16$.

- a. What is the length of $\overline{CX}$?



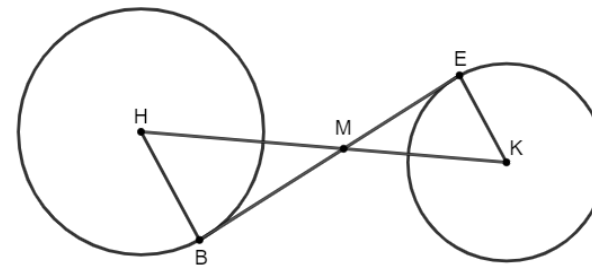
- b. What is the length of $\overline{VC}$?

- c. What is the perimeter of $\triangle VCX$?



12.5.4 Your Turn!

1. Circle H has point B on the circle and circle K has point E on the circle. $\overline{HK}$ and tangent $\overline{EB}$ intersect outside the circles at point M . $m\angle HBM = m\angle MEK = 90^\circ$, $HB = 6$, $BM = 8$, and $MK = 8$.



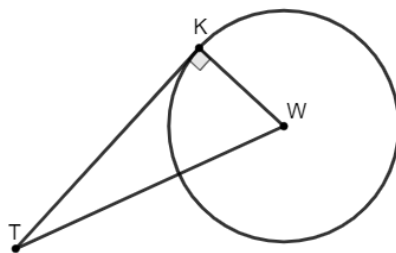
- a. Explain why $\triangle HMB \sim \triangle KME$.

- b. What is the length of $\overline{HK}$?

- c. What is the perimeter of $\triangle KME$?

- d. What is the area of $\triangle KME$?

2. Grace and Zackary are finding the length of tangent $\overline{TK}$ of circle W , where K is a point on the circle. $\overline{TW}$ and tangent $\overline{TK}$ intersect at point T outside the circle. $TW = 18$ and $KW = 7$.

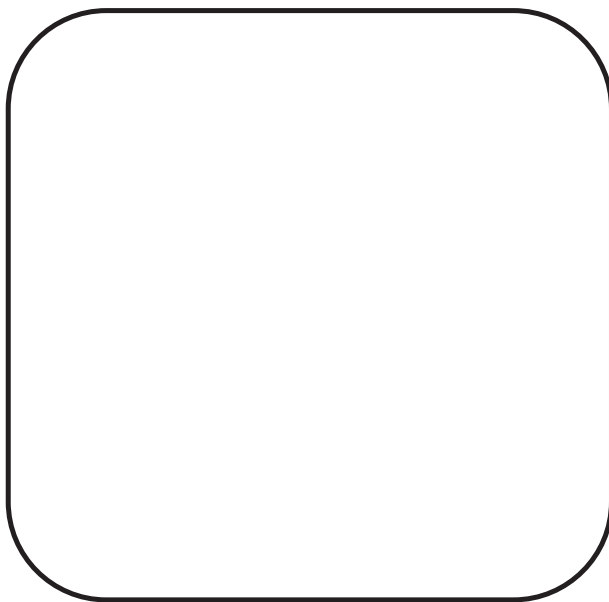


Grace's and Zackary's work is shown.

Grace's Work	Zackary's Work
$7^2 + x^2 = 18^2$	$7^2 + 18^2 = x^2$
$49 + x^2 = 324$	$49 + 324 = x^2$
$x^2 = 275$	$x^2 = 373$
$x = 5\sqrt{11}$	$x = \sqrt{373}$

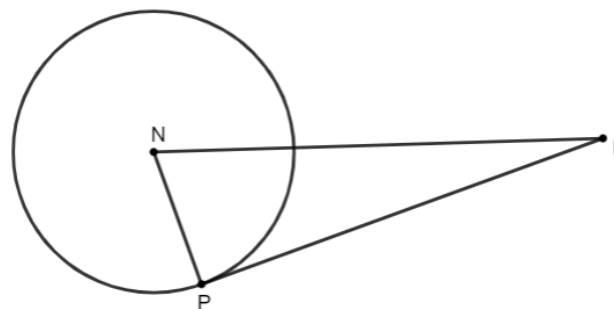
Write a note to the person whose work is incorrect explaining their mistake.

Mmeans
I
Sstart
To
Acquire
Knowledge
Experience
Skills



12.5.5 Wrap-Up: Ticket Out the Door

1. Circle N has point P on the circle and point D outside the circle at the intersection of $\overline{DN}$ and tangent $\overline{DP}$. $NP = 2$ and $DP = 6$.



What is the length of $\overline{ND}$?